RIVER & ATMOSPHERIC POLLUTION BEFORE & AFTER LOCKDOWN

- ADARSH GAWADE

CONTENTS

ATMOSPHERIC POLLUTION







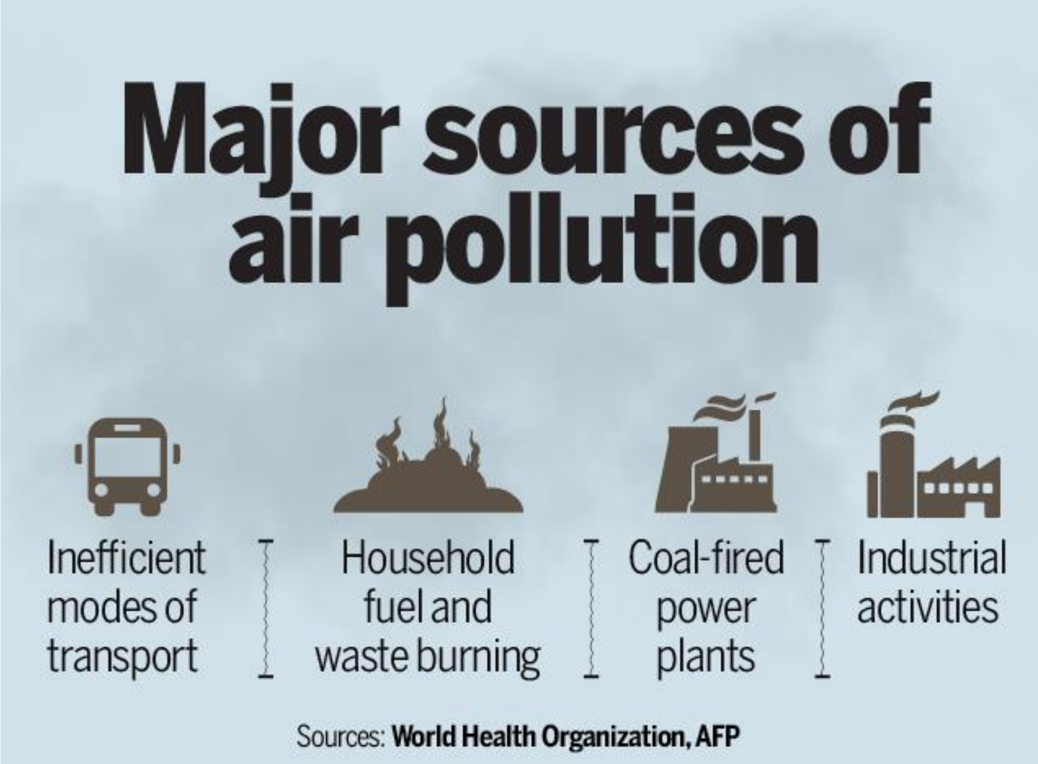


ATMOSPHERIC POLLUTION

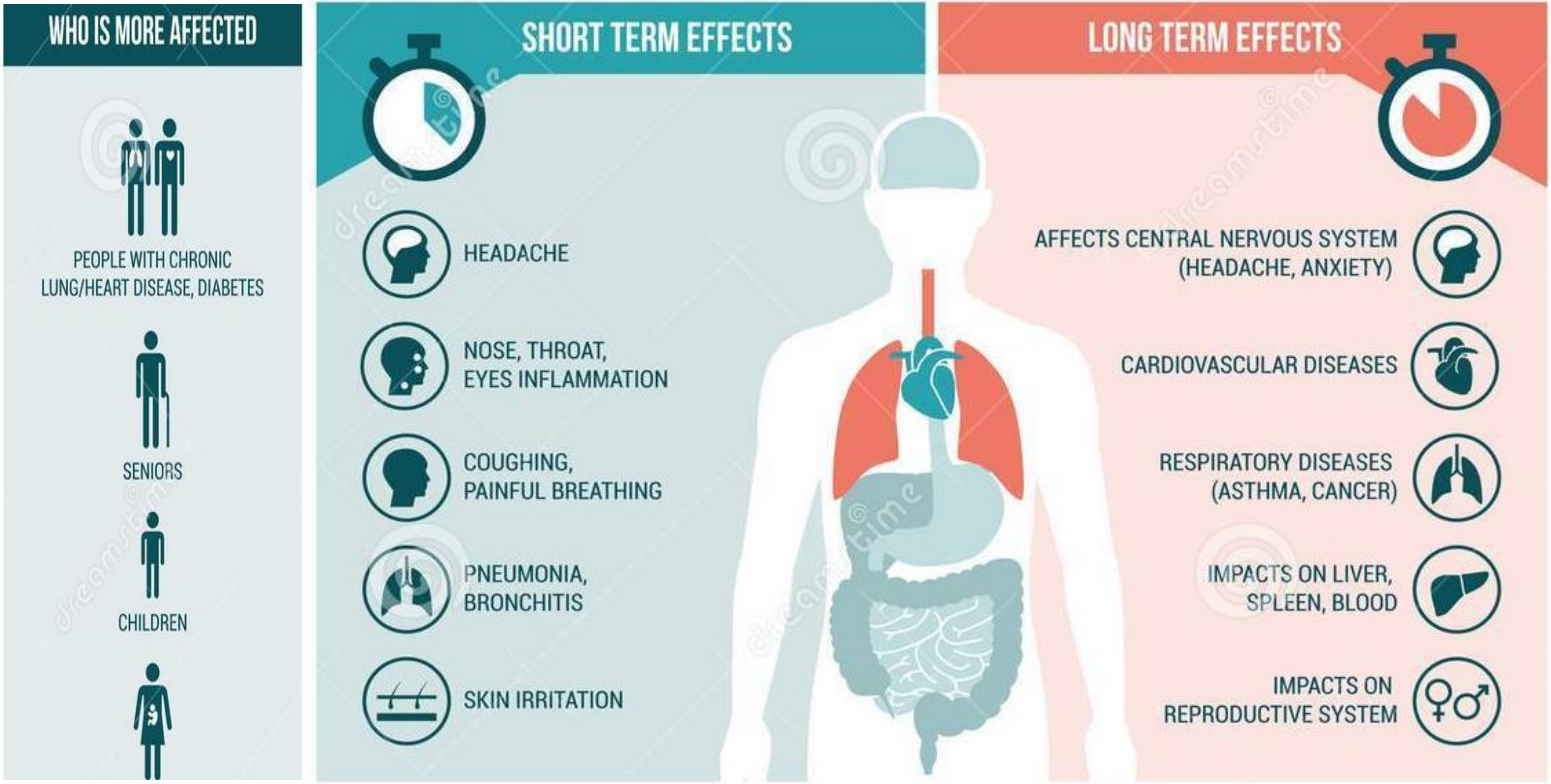
What is Air Pollution ?

Air pollution is a mix of particles and gases that can reach harmful concentrations both outside and indoors. Its effects can range from higher disease risks to rising temperatures. Soot, smoke, mold, pollen, methane, and carbon dioxide are a just few examples of common pollutants.

Air pollution is caused by solid and liquid particles and certain gases that are suspended in the air. These particles and gases can come from car and truck exhaust, factories, dust, pollen, mold spores, volcanoes and wildfires. The solid and liquid particles suspended in our air are called aerosols.



EFFECTS OF AIR POLLUTION ON HEALTH



AIR QUALITY INDEX

- AQI is the air quality index; it gives you the index value that what is the current pollution status in the city, how polluted the air currently is.
- Measurement of AQI
- Eight pollutants namely particulate matter (PM) 10, PM2.5, Ozone (O3), Sulphur dioxide (SO2), nitrogen dioxide (NO2), carbon monoxide (CO), lead (Pb) and ammonia (NH3) act as major parameters in deriving the AQI of an area. Short-term (upto 24-hours and 8-hours for CO and O3) National Ambient Air Quality Standards are prescribed for these pollutants.
- As per CPCB's (Central Pollution Control Board) air quality standards, AQI is categorised into six parts. AQI between 0-50 is considered 'good', 51-100 'satisfactory', 101-200 'moderate', 201-300 'poor', 301-400 'very poor', and between 401-500 'severe'.
- As the AQI value increases, health impacts become serious. For instance, while under satisfactory AQI, sensitive people might witness minor breathing discomfort, severe AQI may cause respiratory impact even on healthy people, and serious health issue in people with existing respiratory issues.

AQI Category (Range)	PM_{10} 24-hr	$PM_{2.5}$ 24-hr	NO_2 24-hr	O_3 8-hr	CO 8-hr (mg/m^3)	SO_2 24-hr	NH_3 24-hr	Pb 24-hr
Good (0-50)	0-50	0-30	0-40	0-50	0-1.0	0-40	0-200	0-0.5
Satisfactory (51-100)	51-100	31-60	41-80	51-100	1.1-2.0	41-80	201-400	0.6 –1.0
Moderate (101-200)	101-250	61-90	81-180	101-168	2.1- 10	81-380	401-800	1.1-2.0
Poor (201-300)	251-350	91-120	181-280	169-208	10.1-17	381-800	801-1200	2.1-3.0
Very poor (301-400)	351-430	121-250	281-400	209-748*	17.1-34	801-1600	1201-1800	3.1-3.5
Severe (401-500)	430 +	250+	400+	748+*	34+	1600+	1800+	3.5+

One hourly monitoring (for mathematical calculation only)

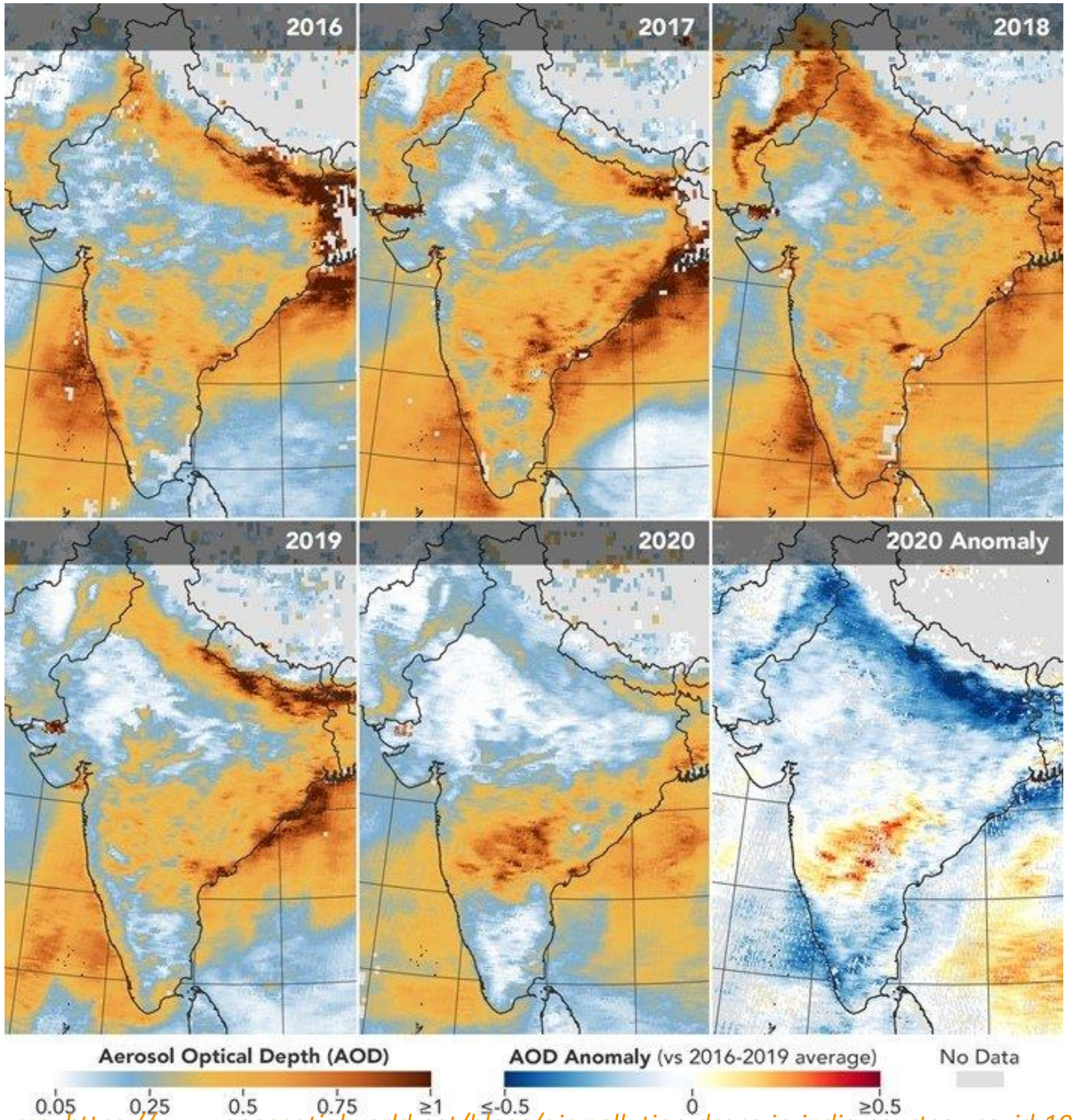
Image credit: National Air Quality Index Report by Central Pollution Control Board

CENTRAL POLLUTION CONTROL BOARD'S AIR QUALITY STANDARDS

AIR QUALITY INDEX (AQI)	CATEGORY
0-50	Good
51-100	Satisfactory
101-200	Moderate
201-300	Poor
301-400	Very Poor
401-500	Severe

BEFORE LOCKDOWN AND AFTER LOCKDOWN AIR POLLUTION STATUS IN INDIA

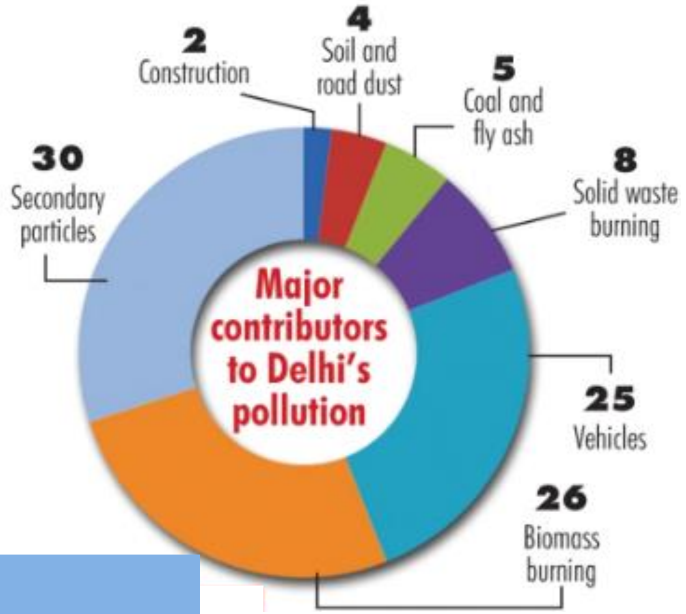
- Air pollution levels have dropped significantly in India owing to the nationwide lockdown imposed in the wake of the Novel Coronavirus outbreak.
- The country has seen a massive dip in vehicular movement and industrial activity, which have resulted in clean and fresh air perhaps in decades.
- NASA satellite sensors have observed aerosol levels at a 20-year low for this time of the year in parts of northern India.
- Aerosols are tiny solid and liquid particles suspended in the air that reduce visibility and can cause damage to human lungs and heart. Around this time every year, aerosols from anthropogenic sources add to air pollution levels in several Indian cities.
- The National Green Tribunal (NGT), India's environment watchdog, has also confirmed that the nationwide lockdown has improved the air quality



Source: https://www.geospatialworld.net/blogs/air-pollution-drops-in-india-courtesy-covid-19/

BEFORE LOCKDOWN AIR POLLUTION STATUS IN DELHI

- **DELHI IS ONE THE MOST POLLUTED CITY IN THE WORLD.**
- Majorly constituted of dust and smoke from stubble burning this grey cover appears unlikely to retreat mainly in the winters.
- Biomass burning accounts for 26% of the pollution source in Delhi. Biomass is burned in the neighbouring states of Haryana and Punjab.
- October to January are seen to be worst months causing impact on health of many residents.
- The AQI can be seen moderate in the months of February and March.



AQI

OCT 2019

Breathing discomfort to most people on prolonged exposure

NOV 2019

Affects healthy people and seriously impacts those with existing diseases

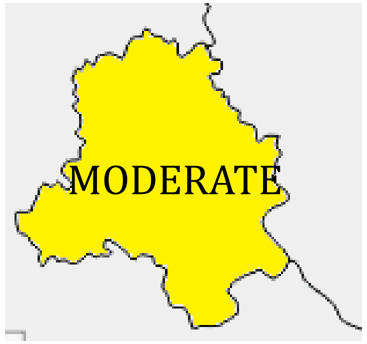
DEC 2019

Breathing discomfort to most people on prolonged exposure

JAN 2020

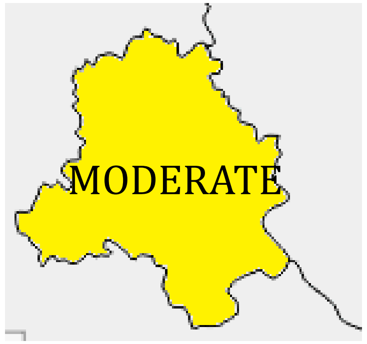
Breathing discomfort to most people on prolonged exposure

FEB 2020



Breathing discomfort to the people with lungs, asthma and heart diseases

MAR 2020

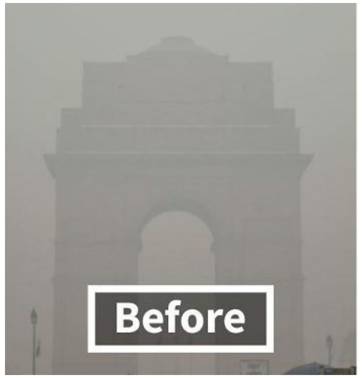


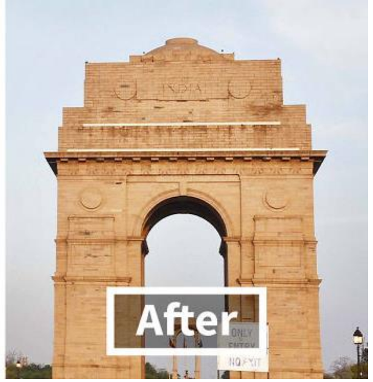
Breathing discomfort to the people with lungs, asthma and heart diseases

AFTER LOCKDOWN AIR POLLUTION STATUS IN DELHI

- Pollution levels in Delhi which had come down by around 79% during the initial phase of the lockdown, mainly owing to no industrial activity, reduced on-road traffic and a pause on construction activities, is on an upswing again as the city gradually opens up.
- In Delhi-NCR, one of the major factors that led to the drop in pollution was a 97% reduction in overall traffic and 91% reduction in trucks and commercial vehicles entering the capital during April, as compared to the pre-lockdown months of December-January.
- AQI can been Moderate in the Months of June and July.

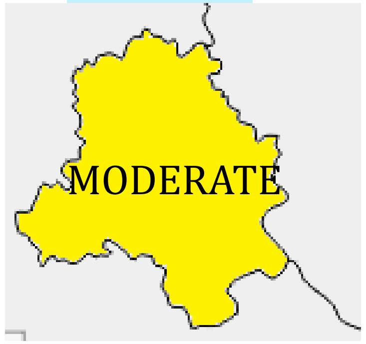
BEFORE LOCKDOWN	AFTER LOCKDOWN





AQI

APR 2020



Breathing discomfort to most people on prolonged exposure

MAY 2020

Minor breathing discomfort to sensitive people

JUN 2020

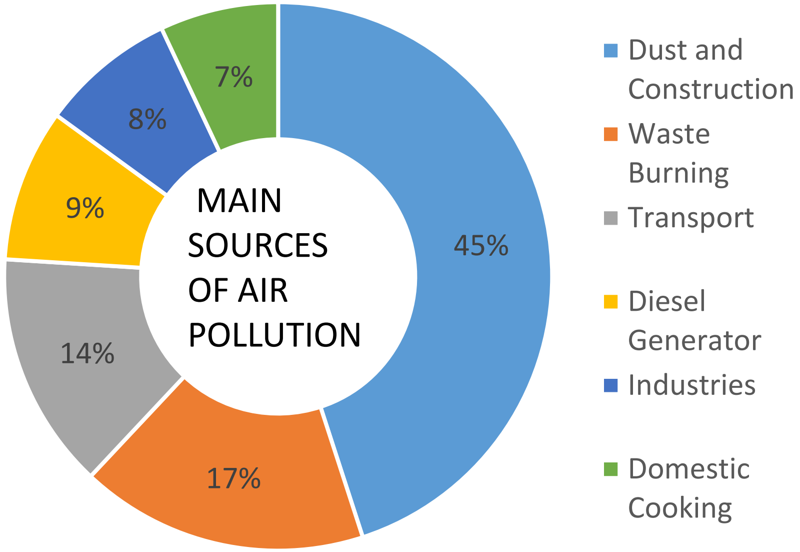
Breathing discomfort to most people on prolonged exposure

JUL 2020

Breathing discomfort to most people on prolonged exposure

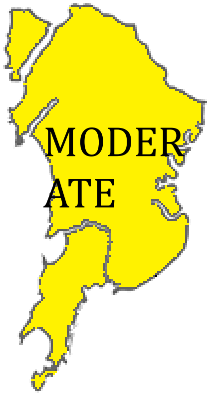
BEFORE LOCKDOWN AIR POLLUTION STATUS IN MUMBAI

- Air Pollution in the City of Mumbai is seen moderate before lockdown.
- Road and construction dust is the major source of pollution in Mumbai. i.e the highest percentage of PM(particulate matter) comes from road and construction dust.
- The increasing number of private vehicles over public transport would replace the industrial fumes by transport emissions as the prime source of air pollution in the city
- Followed by the power plants which contribute of the total PM levels.
- As per the reports from an air quality assessment conducted by CPCB and NEERI, the emissions from heavy-duty vehicles running on diesel were also found to be majorly contributing to the pollution.

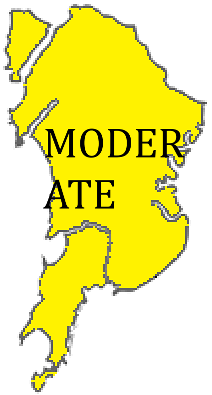


AQI

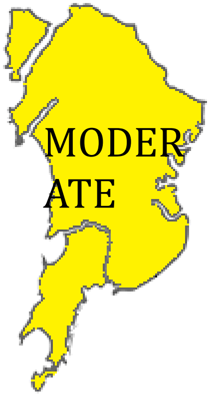
OCT 2019



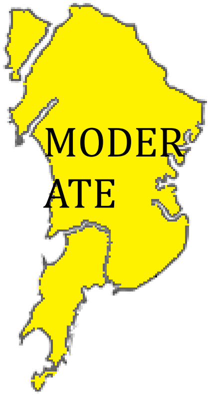
NOV 2019



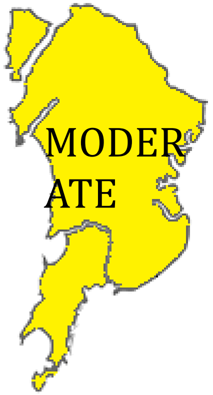
DEC 2019



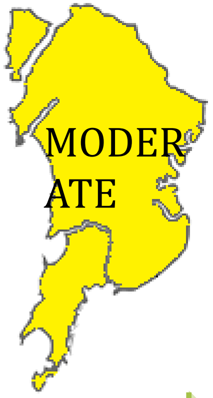
JAN 2020



FEB 2020



MAR 2020



Breathing discomfort to the people with lungs, asthma and heart diseases

AFTER LOCKDOWN AIR POLLUTION STATUS IN MUMBAI

- The city of Mumbai has registered a spell of clean air after the commencement of lockdown on March 25 in view of coronavirus pandemic.
- The city witnessed a 76 per cent decline NO2 levels, while PM2.5 saw a decline of 54 per cent compared to the same period last year.
- The lockdown period helped us understand the effects of anthropogenic (human-generated) emissions to our environment.
- This drastic fall in pollution levels is that out of the eight primary polluting sources, four were completely absent during the lockdown — vehicular emissions, emissions from industry, construction and brick kilns.

AQI

APR 2020

Minor breathing discomfort to sensitive people

MAY 2020

Minimal impact

JUN 2020

Minimal impact

JUL 2020

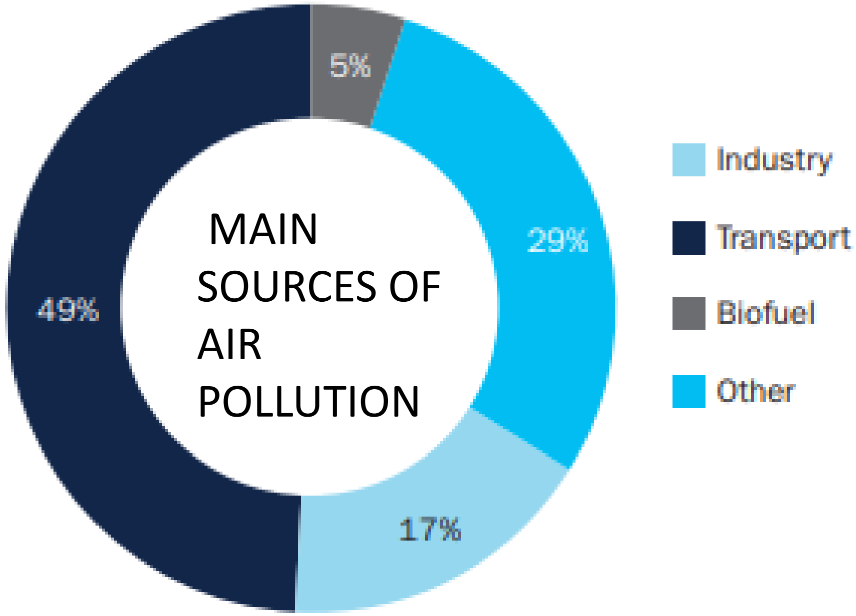
Minimal impact

BEFORE LOCKDOWN

AFTER LOCKDOWN

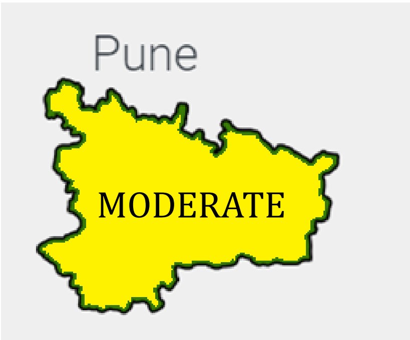
BEFORE LOCKDOWN AIR POLLUTION STATUS IN PUNE

- Over the past five years, the air quality in the Indian city of Pune has deteriorated significantly. Data collected by the Indian Institute of Tropical Meteorology (IITM) indicates that 2018 was reportedly the second most polluted year since 2013.
- The key results from Pune's emissions inventory for fine particles indicate that the transportation sector was responsible for almost half of primary emissions, the air pollution emitted directly from combustion or other sources. The other contributors were suspended dust (29%), industrial operations (17%), and solid fuel combustion (5%).
- It is estimated that estimated 733 excess deaths per million people in Pune due to cardiovascular diseases from air pollution exposures (PM10 and SO2) each year.

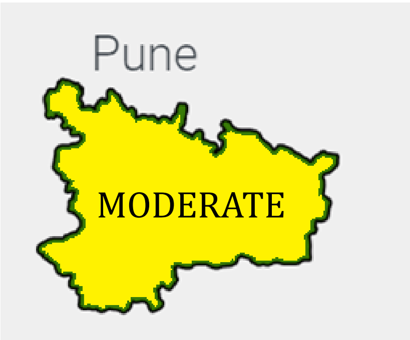


AQI

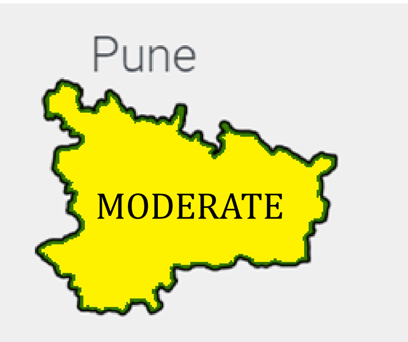
OCT 2019



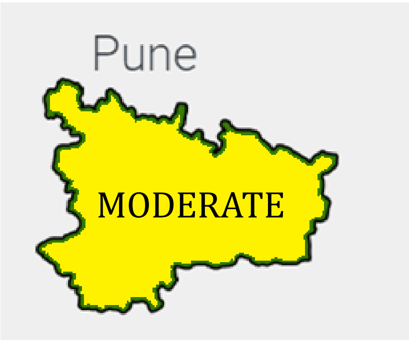
NOV 2019



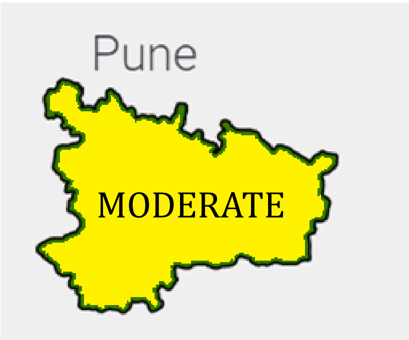
DEC 2019



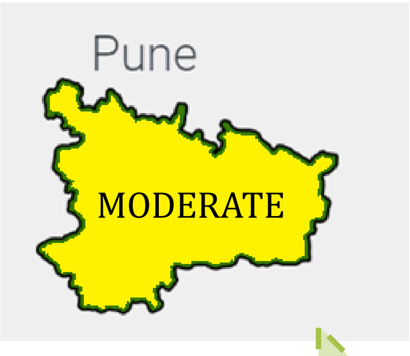
JAN 2020



FEB 2020



MAR 2020



Breathing discomfort to the people with lungs, asthma and heart diseases

AFTER LOCKDOWN AIR POLLUTION STATUS IN PUNE

- As the number of vehicles on the road has come down in the city during lockdown, the air quality has improved drastically.
- The level of Nitrogen Oxide (NOx) pollution, which can increase the risk of respiratory conditions, has also reduced. NOx pollution is mainly caused due to a high motor vehicle traffic. In Pune, NOx pollution has reduced by 43%
- Before lockdown the AQI of Pune was Moderate and after lockdown the AQI is seen as Satisfactory.
- Local factors like shutting down of industries and construction and traffic have contributed in improving the air quality. Rain is also helping, but the curbs on local emissions are playing a significant role.

AQI

APR 2020

SATISFA CTORY

MAY 2020

SATISFA CTORY

JUN 2020

SATISFA CTORY

JUL 2020

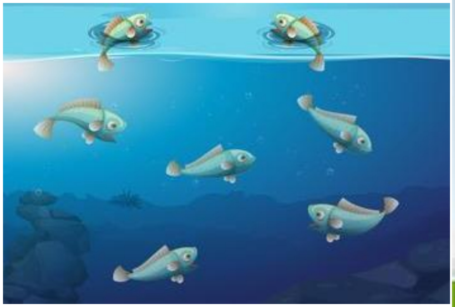
SATISFA CTORY

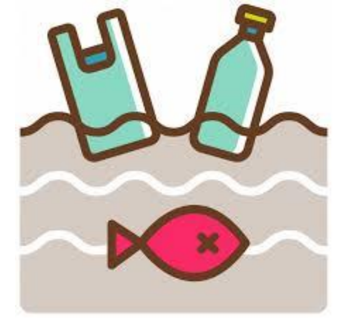
Minor breathing discomfort to sensitive people

BEFORE LOCKDOWN

AFTER LOCKDOWN

RIVER POLLUTION



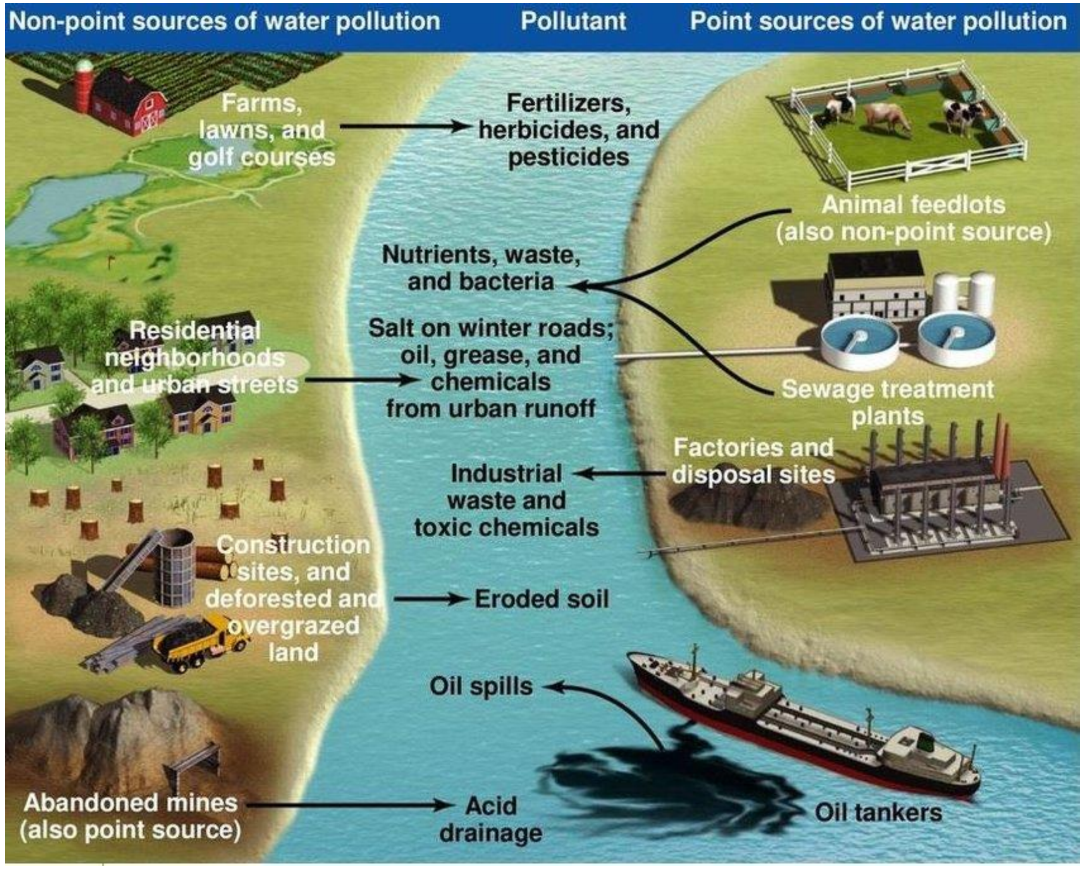


RIVER POLLUTION

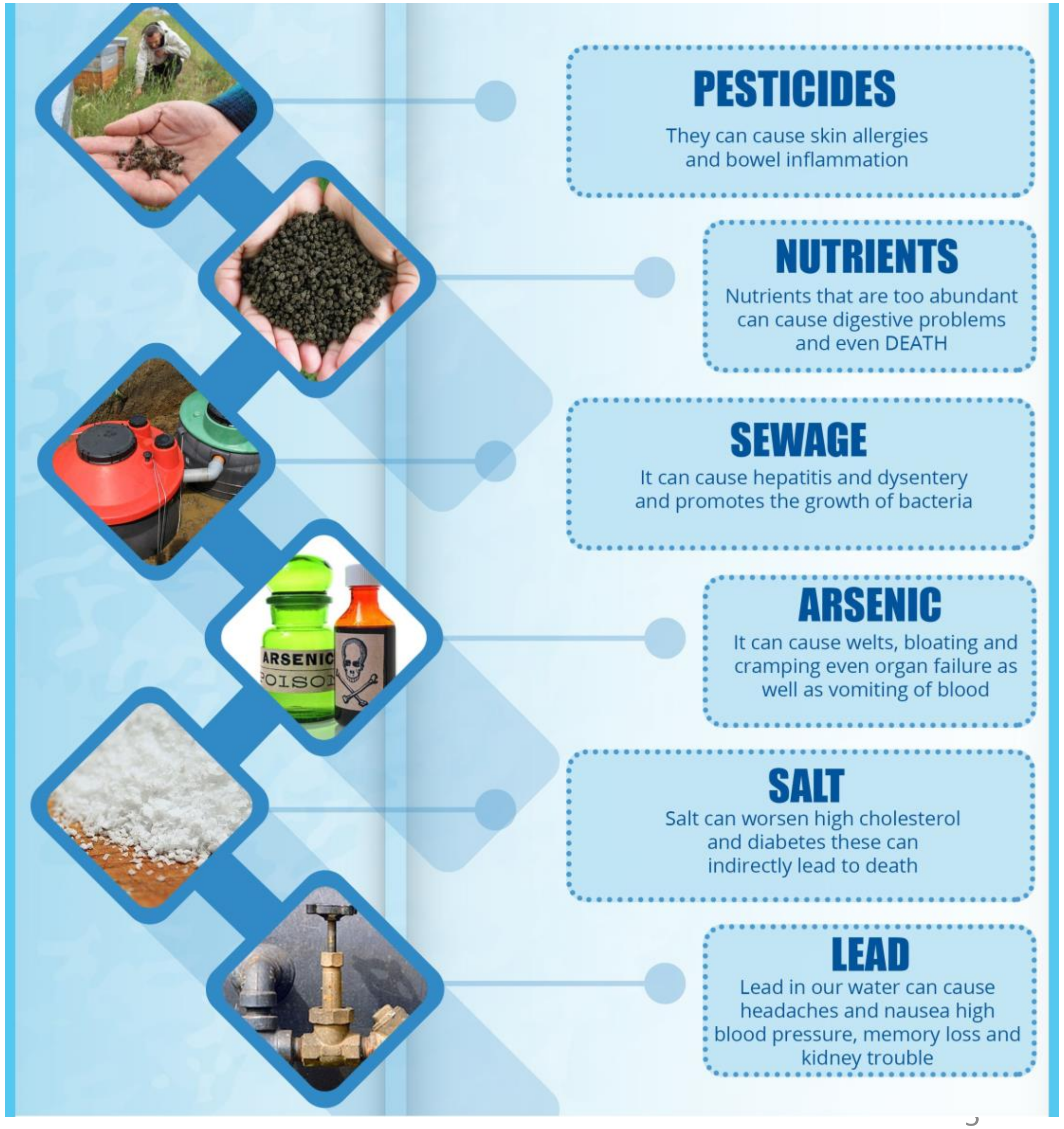
River Pollution

- River pollution is the contamination of water in the river usually caused due to human activities. Water pollution is any change in the physical, chemical or biological properties of water that will have a detrimental consequence of any living organism.

CAUSES of River Pollution

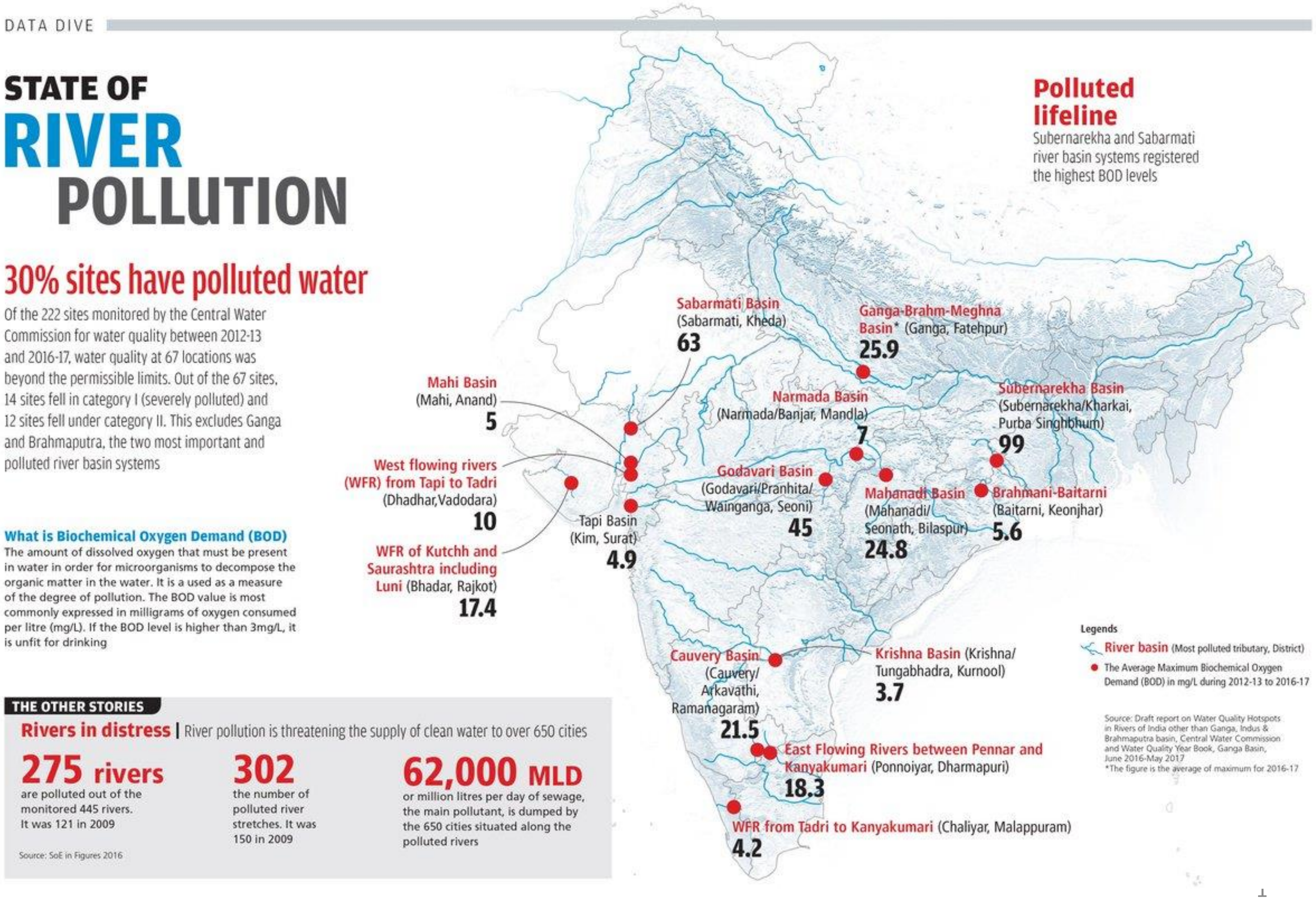


EFFECTS of River Pollution



RIVER POLLUTION STATUS IN INDIA

- Based on the recommendations of the **National Green Tribunal**, the CPCB apprised the **States of the extent of pollution in their rivers**.
- The number of **polluted stretches in India's rivers** has **increased to 351 from 302** two years ago.
- The number of **critically polluted stretches** where water quality indicators are the poorest has gone up to 45 from 34, according to an assessment by the **Central Pollution Control Board (CPCB).**
- The **Rs. 20,000 crore clean-up of the Ganga** may be the most visible of the government's efforts to tackle pollution, the CPCB says several of the river's stretches in **Bihar and Uttar Pradesh** are actually far less polluted than many rivers in **Maharashtra, Assam and Gujarat.**

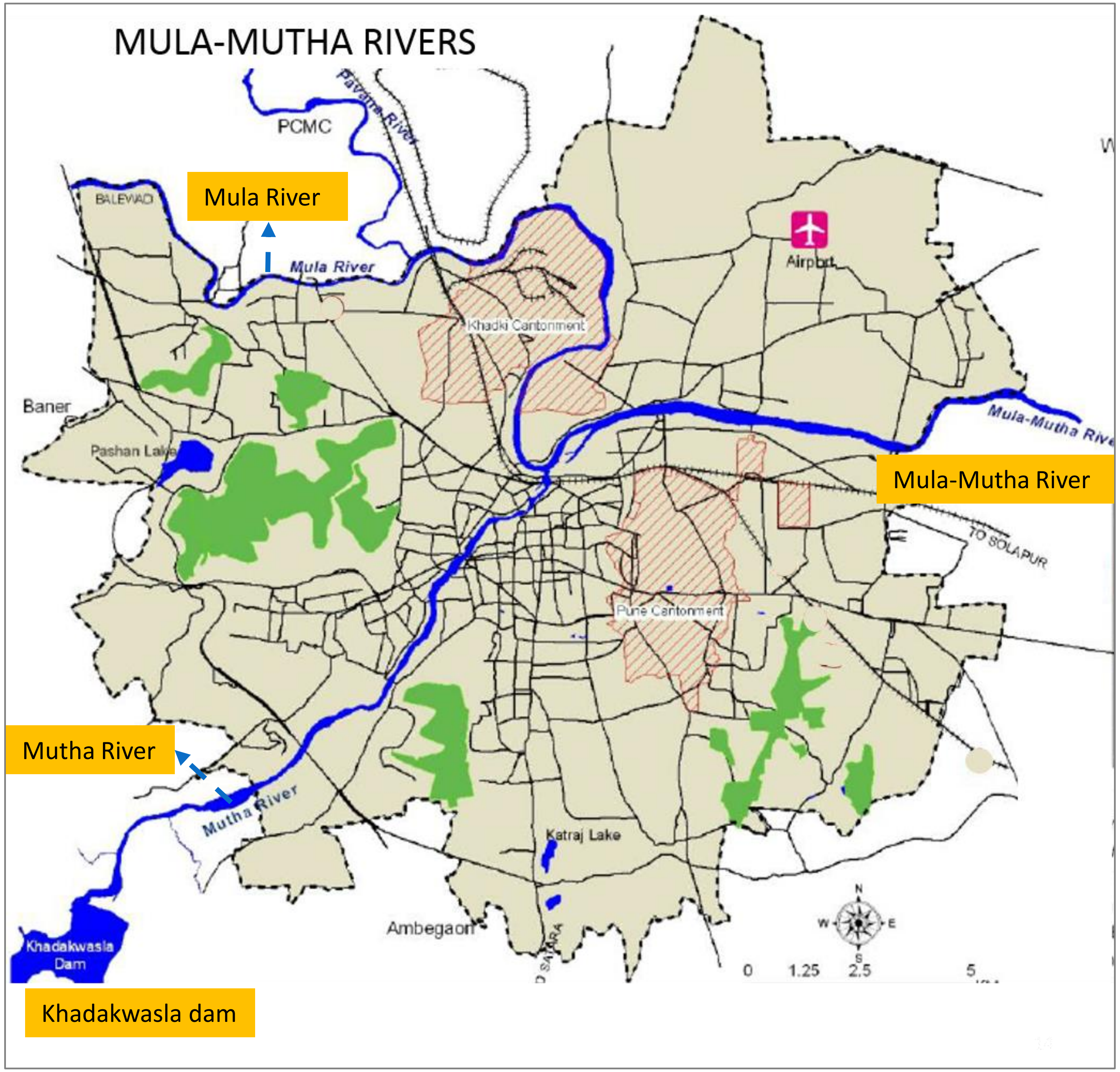


MULA-MUTHA RIVER, PUNE

- Pune lies at the confluence of Mula and Mutha rivers. Mutha river origin from Khadakwasla dam & it passes through Dhari, Nanded, z-bridge, Junabazzar, Pune RTO and ends at Sangamwadi. Both Mula and Mutha River merged at Sangamwadi which is further joined by Indrayani & Bhima River.
- Due to rapid development of city the pollution load into river Mula - Mutha has been increased as this river is passing from major areas of Pune.
- Industrial areas having industries like Hindusthan antibiotics, Teleco, Bajaj Auto, paper mills and others hundreds of small scale industries. It is among the 35 polluted river stretches in the country, classified as Priority 1 (the highest risk category) by the Central Pollution Control Board (CPCB).

The major reasons for pollution of Mula-Mutha are:

- Discharge of untreated domestic waste-water into the river due to inadequate sewerage system (including pumping stations)
- Sewage treatment capacity in the town
- Open defecation on the river banks.
- Dumping of Solid Waste



MULA-MUTHA RIVER, PUNE

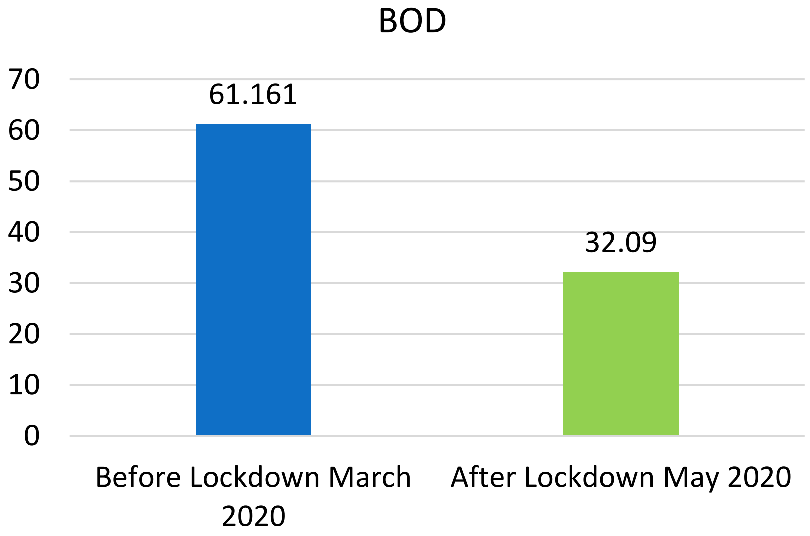
BEFORE LOCKDOWN AND POST LOCKDOWN variations can be seen in the water quality of the rivers. There are three parameters that show the water quality in a river.

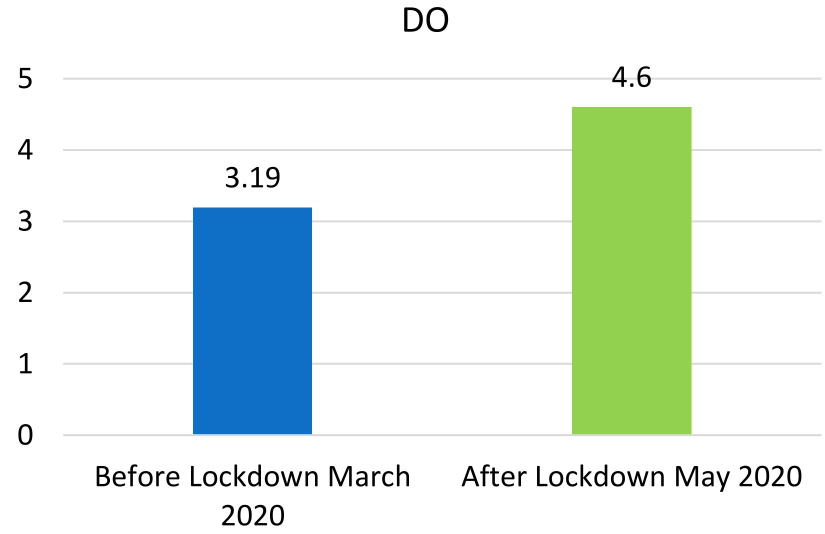
- Biochemical oxygen demand (BOD)
- Chemical oxygen demand (COD)
- Dissolved oxygen (DO)

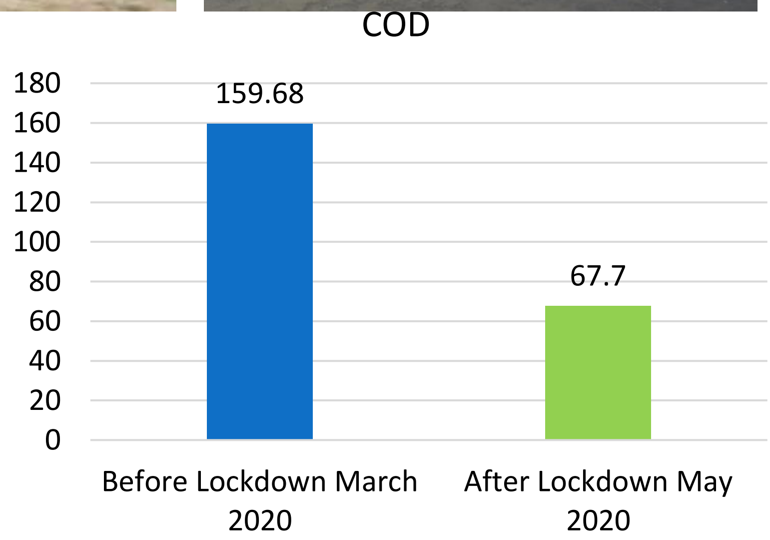
In all the three parameters there is a drastic improvement and BOD and COD have come down below the CPCB parameters. The main reason for this is commercial establishments, offices, hotels have closed down so the sewage generation has reduced.

BEFORE LOCKDOWN

AFTER LOCKDOWN

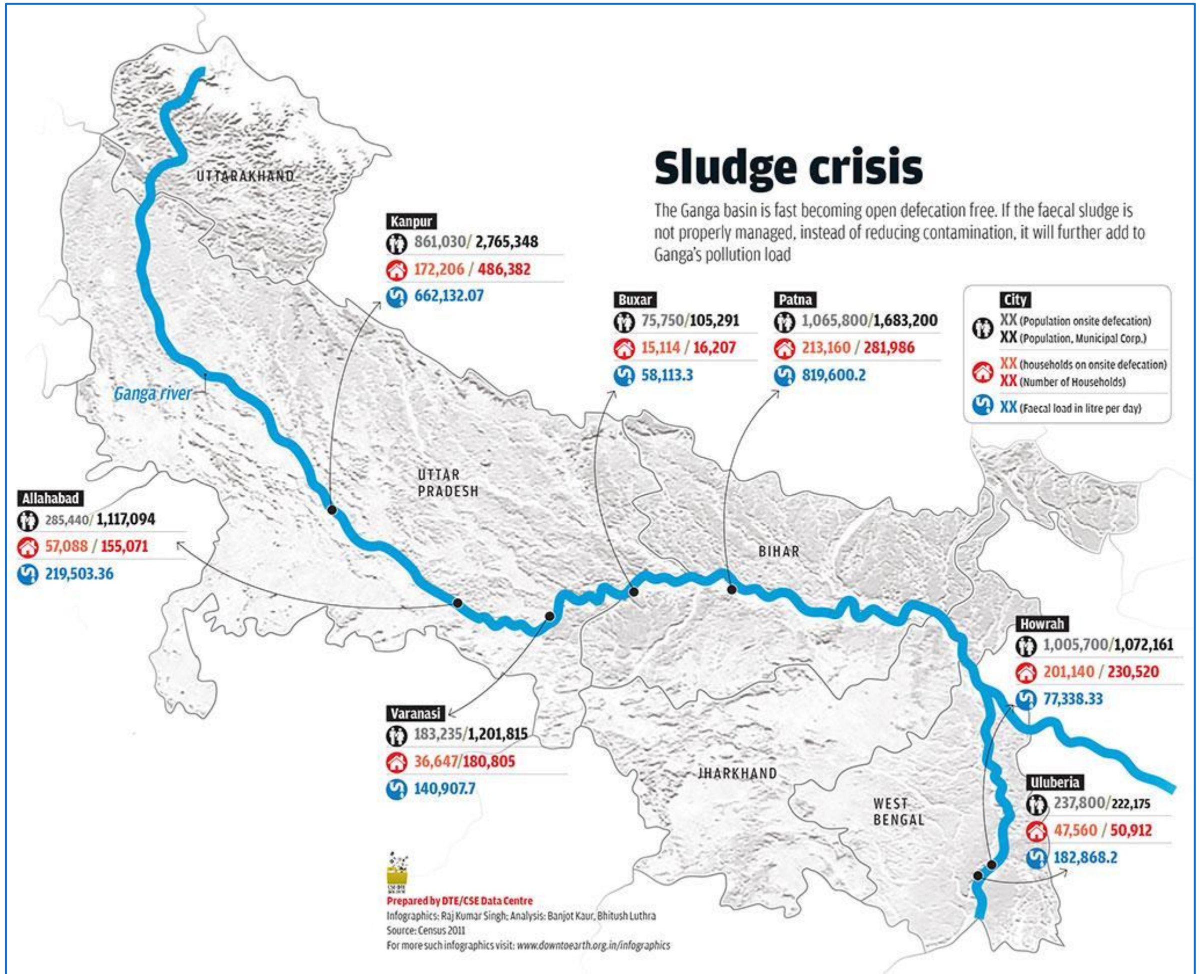






GANGA RIVER

- Ganges River, Hindi Ganga, great river of the plains of the northern Indian subcontinent.
- Although officially as well as popularly called the Ganga in Hindi and in other Indian languages, internationally it is known by its conventional name, the Ganges
- Rising in the Himalayas and emptying into the Bay of Bengal, it drains one-fourth of the territory of India, and its basin supports hundreds of millions of people. The greater part of the Indo-Gangetic Plain, across which it flows, is the heartland of the region known as Hindustan.
- Causes of Pollution in Ganges
  - Discharge of untreated sewage
  - Discharge of industrial waste
  - Agricultural runoff
  - Dumping of Solid waste



GANGA RIVER

- A Central Pollution Control Board (CPCB) report attributed the drop in pollution to reduced release of industrial waste and reduced discharges/dumping of wastes from various non-point sources in the river stream.
- The Ganga water quality improved remarkably during the lockdown period. The 2,500-kilometre river has been an important part of India’s history, identity, religious beliefs and economy for thousands of years.
- According to the real-time water monitoring data of the CPCB, out of the 36 monitoring units placed at various points of the Ganga river, the water quality around 27 points was found suitable for bathing and propagation of wildlife and fisheries.
- More than 80 per cent of pollution in the Ganga is due to domestic sewage from surrounding towns and villages. The rest is contributed by industrial waste.
- During the lockdown, domestic sewage would have increased owing to increased demand for water to maintain hand-washing hygiene. Industrial waste, however, stopped entering the Ganga.
- Other activities such as tourism, fairs, bathing and cloth washing near the Ghats were curtailed.

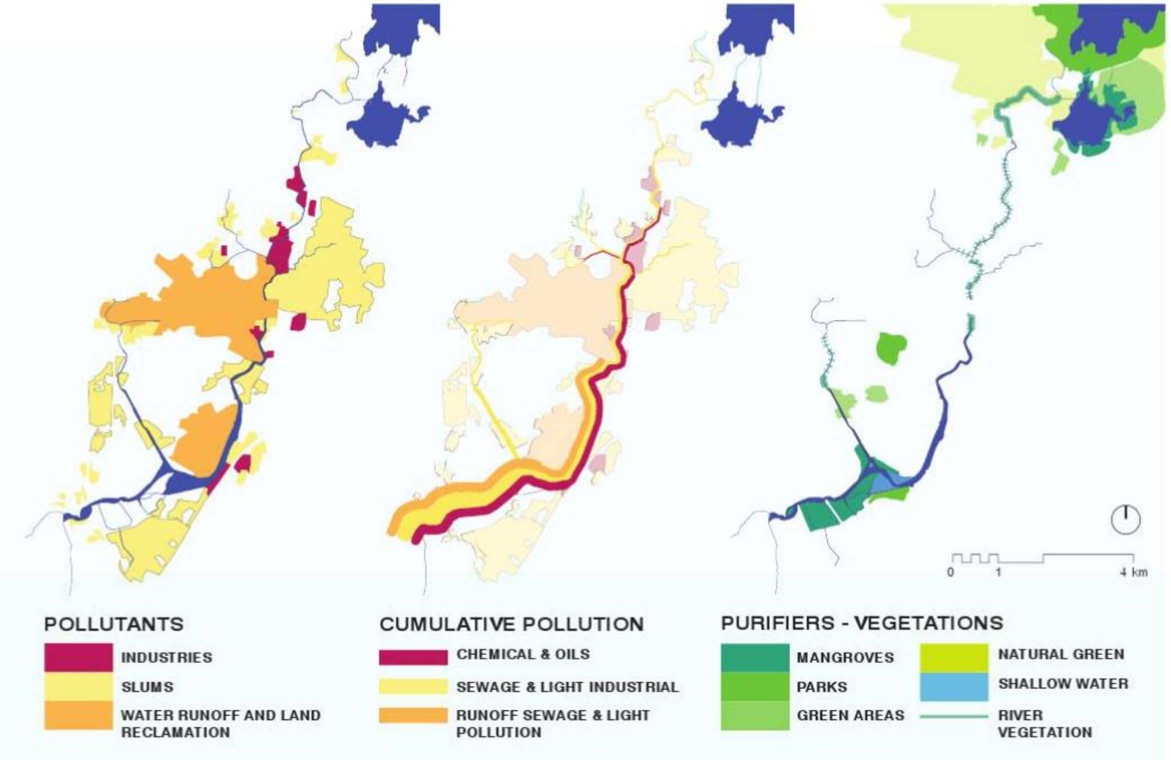
BEFORE LOCKDOWN

AFTER LOCKDOWN

MITHI RIVER, MUMBAI

- The Mithi originates in the hills to the east of the Sanjay Gandhi National Park.
- The Mithi River flows 17.8 km before draining at Mahim Bay and into the Arabian Sea. Its journey includes cramped slums and squatters at Morarji Nagar, Bhim Nagar, Dharavi, and the industrial and residential complexes of Powai, Saki Naka, Kurla and Mahim. Morarji Nagar constitutes its north end.
- One of the oldest rivers in the state, the Mithi is now nothing more than a sewage drain. In July 2019, Maharashtra Environment Minister Ramdas Kadam declared that the stream consisted of 93% domestic sewage and 7% industrial waste. The Mithi is a critical storm water drain for Mumbai.
- Causes of Mithi River Pollution
  - Discharge of untreated sewage
  - Discharge of industrial waste
  - Dumping of Solid waste
- Clogged with plastic, garbage and sewage, the river couldn't bear the load. Instead of flowing downstream into the Arabian Sea, the floodwaters seeped onto the streets of the city, along with gallons of waste.

EXISTING CONDITIONS OF MITHI RIVER

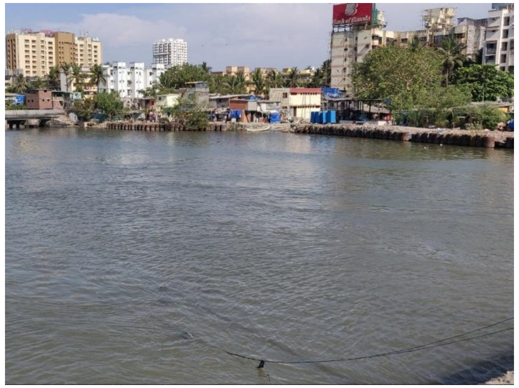


MITHI RIVER, MUMBAI

- Water quality at major beaches in the city witnessed an improvement during the lockdown. However, the Mithi river was more polluted than in pre-lockdown months and the corresponding period last year.
- In april, the Mithi and Mahim creek, into which the river flows, were the only two listed water bodies in the 'heavily polluted' category with pollution levels at 25.7 and 37.4 (red category). Versova, Juhu, Nariman Point, Gateway of India, Malabar Hill, Girgaum Chowpatty, Haji Ali, Shivaji Park Dadar and Worli Sea Face all recorded levels ranging from 51 and 58 (yellow category or non-polluted) in April.
- During January and February 2020 as well as in March and April 2019, Mahim creek was in the orange category while Mithi was in the red category.
- There was an overall reduction in sewage load at marine outfalls.
- There was a reduction in untreated domestic sewage coming from large hotels, offices, industries and activities from other establishments that were all shut during the first phase of the lockdown. Also, existing sewage treatment plants (STPs) functioned at optimal capacity along marine outfalls to treat waste efficiently, which was only coming from slums and households

BEFORE LOCKDOWN

AFTER LOCKDOWN



LOCKDOWN HAS PROVED A **VENTILATOR** FOR NATURE.

WE NEED TO GO GREEN AND PROTECT NATURE AS THERE IS NO PLANET "B"

THANK YOU!